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EUROPEAN PATENT OFFICE
D-80298 MÜNCHEN
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By EPOLINE

Paris, April 19, 2023

Our Ref.: B21101893EP

Re: European Patent Application **No. 20776930.8**
Regional phase of PCT/KR2020/003809
Filed March 19, 2020
in the name of **LG Innotek Co., Ltd.**

Sirs,

The Applicant wishes to proceed further with the above-mentioned European patent application.

This letter further implies an authorization to the European Patent Office, to debit the account No. 28040012 for any fee which may be due for completing the examination/designation steps and proceeding further with the application.

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In response to the European Search Opinion (ESOP):

- new pages 75-81 are filed to replace the set of claims previously on file;
- amended pages 1-74 of the description are filed to replace the previous pages 1-69;
- the following comments are submitted.

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A marked-up copy of the amended pages is enclosed to facilitate the examiner's review.

¤ ¤ ¤

The claims have been amended to address the clarity objections formed in § 1.1 to 1.4 of the ESOP.

The introduction of the invention is also amended to disclose the invention as claimed, under Rule 42(1)c) EPC (§ 2.3 of the ESOP).

In § 2.4 of the ESOP, the Examiner suggests that the independent claim should be drafted in two-part form in accordance with Rule 43(1) EPC. The Applicant notes that Rule 43(1) requires the two-part form for an independent claim only where appropriate. In view of the nature of the present invention and its relationship with the cited art, we do not believe that intelligibility of the claim would be improved by using the two-part form. Hence, a one-part form is retained at this stage.

The background section of the description is amended to cite the documents D1 (EP 1 670 069 A1), D2 (DE 10 2011 002340 A1) and D3 (US 2007/030676 A1) reflecting the cited art, under Rule 42(1)b) EPC (§ 2.5 of the ESOP).

Reference signs in parentheses have been introduced into the claims where appropriate as suggested in § 2.6 of the ESOP (Rule 43(7) EPC).

Referring to the above indications provided under Rule 137(4) EPC, it is submitted that the amendments to the application comply with Article 123(2) EPC and that the claims are clear under Article 84 EPC.

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The Applicant submits that claim 1, at least as amended hereby, defines patentable subject matter over the cited art.

None of the reference disclose a lighting device where "*the plurality of light sources includes a first light source and a second light source adjacent to each other,*
wherein a virtual curve passes through centers of the plurality of light sources,
wherein each of the light sources includes a light emitting device,
wherein the plurality of convex portions includes a first convex portion facing the first light source and a second convex portion facing the second light source, a convex surface of the first and second convex portions has a curvature, and a virtual circle is formed by each of the first and second convex portions,
wherein a first virtual straight line passes through a center of the virtual circle formed by the first convex portion and a first point where a center of the first light source is in contact with the virtual curve, and a second virtual straight line passes through a center of the virtual circle formed by the second convex portion and a second point where a center of the second light source is in contact with the virtual curve, wherein the first virtual straight line and the second virtual straight line are parallel to each other,

wherein an angle between a first tangent line contacting the virtual curve at the first point and the first virtual straight line includes a first obtuse angle."
Thanks to the particular claimed arrangement, it is possible to improve the uniformity of light along the light exit direction.

A technical problem to solve may be how to improve optical reliability?

Thus, the subject matter of claim 1 involves an inventive step over any of the cited art.

Claims 2-13 are linked to an independent claim which is considered to fulfil the novelty and inventive step requirements. At least for this reason, claims 2-13 define patentable subject matter.

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The application is believed to be in condition for allowance. Issuance of a Communication under Rule 71(3) is requested.

In the event the Examiner has any doubts about the allowability of the application, he is kindly invited to call the undersigned at his convenience to discuss any outstanding issues.

As a precaution, oral proceedings under Article 116 EPC are requested prior to any decision to reject the application. If we are summoned to attend oral proceedings for this application, we further request that they take place as a videoconference.

Yours very truly,

/Laurence LOUMES/
PLASSERAUD IP
Association of Representatives No. 280

Encls. - New pages 1-81
- Marked-up copy of amended pages